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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 01/11/2005 |
| **PI Name:** | McNeil, Paul L Ph.D. | | |
| **Project Title:** | Cellular and Molecular Mechanisms Mediating Rapid Resealing of Gravity-Induced Plasma Membrane Disruptions | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Cell Culture | | |
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| **City:** | Athens | **State:** | GA |
| **Zip Code:** | 30912 | **Congressional District:** | 12 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Space Biology 03-OBPR-03 |
| **Start Date:** | 02/01/2004 | **End Date:** | 09/30/2006 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNA04CC44G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The major goals of the research program remain unchanged: 1) to elucidate the organelle and molecular components used by skeletal muscle to repair tears in their surface plasma membrane; 2) to identify cell and molecular events leading to muscle hypertrophy | | |
| **Rationale for HRP Directed Research:** | | | |
| **Research Impact/Earth Benefits:** | Understanding of muscle disease. | | |

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| **Task Progress:** | We have again during this funding period focused on the first of these goals, understanding repair. Listed below are two major areas of advance.<br>o Testing the role of lysosomes in repair Resealing after wounding, the process of repair following plasma membrane damage, requires exocytosis. Vacuolins are molecules that induce rapid formation of large, swollen structures derived from endosomes and lysosomes by homotypic fusion combined with uncontrolled fusion of the inner and limiting membranes of these organelles. Vacuolin-1, the most potent compound, blocks the Ca(2+)-dependent exocytosis of lysosomes induced by ionomycin or plasma membrane wounding, without affecting the process of resealing. In contrast, other cell structures and membrane trafficking functions including exocytosis of enlargeosomes are unaffected. Because cells heal normally in the presence of vacuolin-1, we suggest that lysosomes are dispensable for resealing. This work in now published (EMBO rep., see below).<br><br>o Developing methods for chemical and genetic screening We have hypothesized in an invited review (McNeil and Kirchhausen, see below) that resealing employs novel organelle and protein components. One important pathway to the discovery of these components is a genetic or chemical screen. Such screens require that the cell process under study, resealing in this case, can be assessed quantitatively in multi-well dishes. We have been focusing on developing such an assay, and can now report some progress on this front. Cultured human (Hela) or drosophila (S2) cells present in multi-well dishes are subject to shear stress (tituration) in the presence of Ca2+ (resealing permissive condition) or the absence of Ca2+ (resealing non-permissive condition). Then propidium iodide is added. This dye enters and fluorescently stains the nuclei of dead cells only. If cells fail to reseal the disruptions induced by shear stress, they become labeled with the dye and the resultant increase fluorescence signal can be measured in a fluorescence multi-well plate reader. The prediction is that this fluorescence signal should be higher in cells wounded in the absence of Ca2+ then in cell wounded in its presence. Our preliminary experiments (see graph below) confirm this prediction. Importantly, all of the maneuvers required by this assay can be performed by the robotics associated with chemical and genetic screening facilities, we are optimistic that we will be able to conduct fruitful screens in the near future, enabling us to identify novel organelle and protein components of the resealing machinery. This could lead to major advances in the field. |
| **Bibliography Type:** | Description: (Last Updated: 06/11/2008) |
| **Articles in Peer-reviewed Journals** | Cerny J, Feng Y, Yu A, Miyake K, Borgonovo B, Klumperman J, Meldolesi J, McNeil PL, Kirchhausen T. "The small chemical vacuolin-1 inhibits Ca(2+)-dependent lysosomal exocytosis but not cell resealing." EMBO Rep. 2004 Sep;5(9):883-8. PMID: 15332114 , Sep-2004 |
| **Articles in Peer-reviewed Journals** | McNeil PL, Kirchhausen T. "An emergency response team for membrane repair." submitted, March 2005. , Mar-2005 |
| **Presentation** | McNeil, P.L. "How cells repair membranes" Lysosomes Gordon Reseach Conference<br>May-2004 |
| **Presentation** | McNeil, P.L. "How cells repair membranes, and why" Membrane Repair. VII Workshop on Pore Forming Toxins. October, 2004, Mainz, Germany<br>Oct-2004 |
| **Presentation** | McNeil, P.L. "How cells repair surface tears and what happens when they fail" Instituto Gulbenkan de Ciencia, Portugal<br>May-2004 |
| **Presentation** | McNeil, P.L. "How cells repair surface tears and what happens when they fail" Rockerfeller University, NYC<br>Jul-2004 |